



FACULTY OF COMPUTING & INFORMATICS (FCI)
CCP6224 OBJECT-ORIENTED ANALYSIS & DESIGN
ASSIGNMENT

TUTORIAL SECTION: T12L

GROUP NAME: GROUP 03

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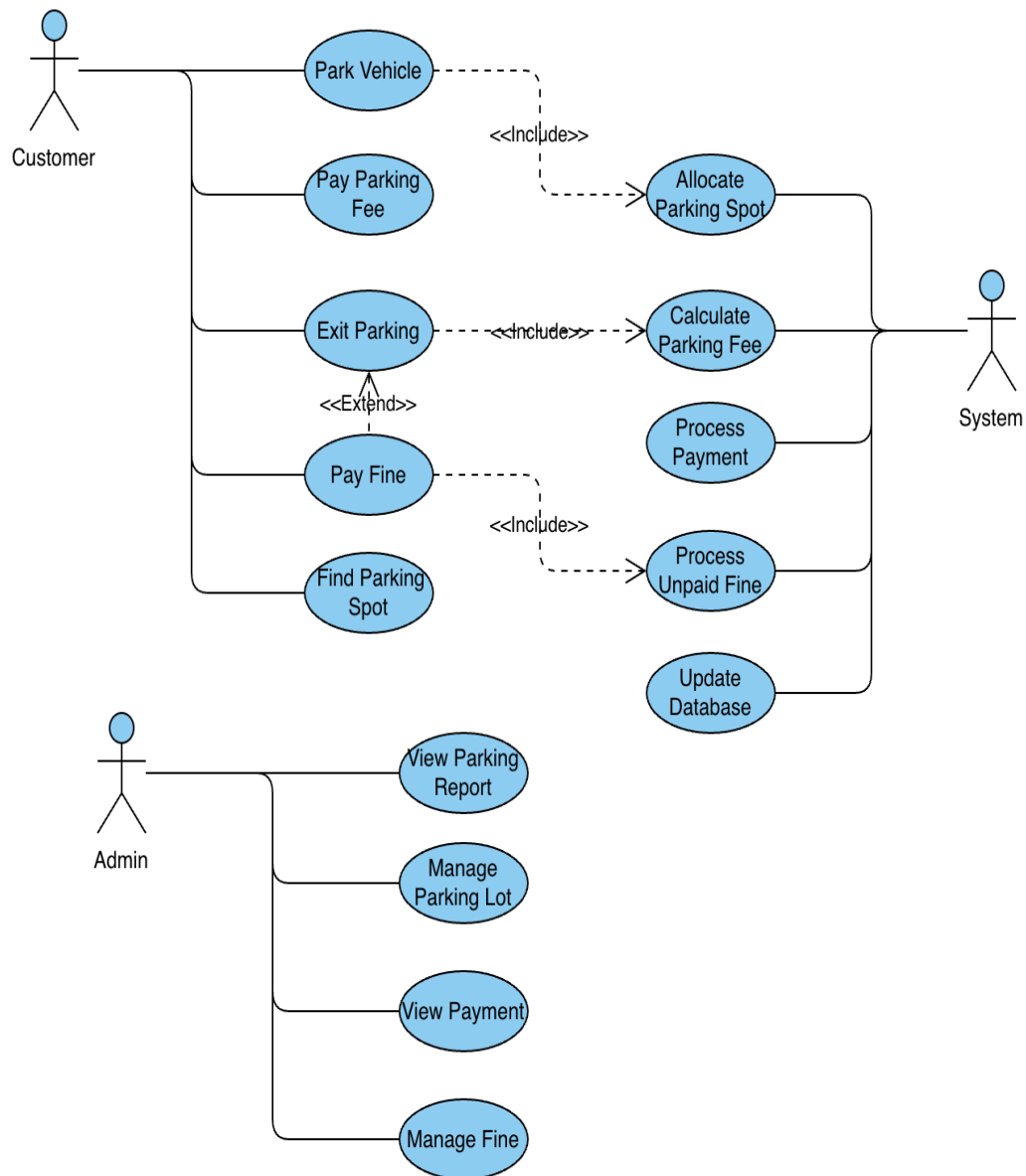
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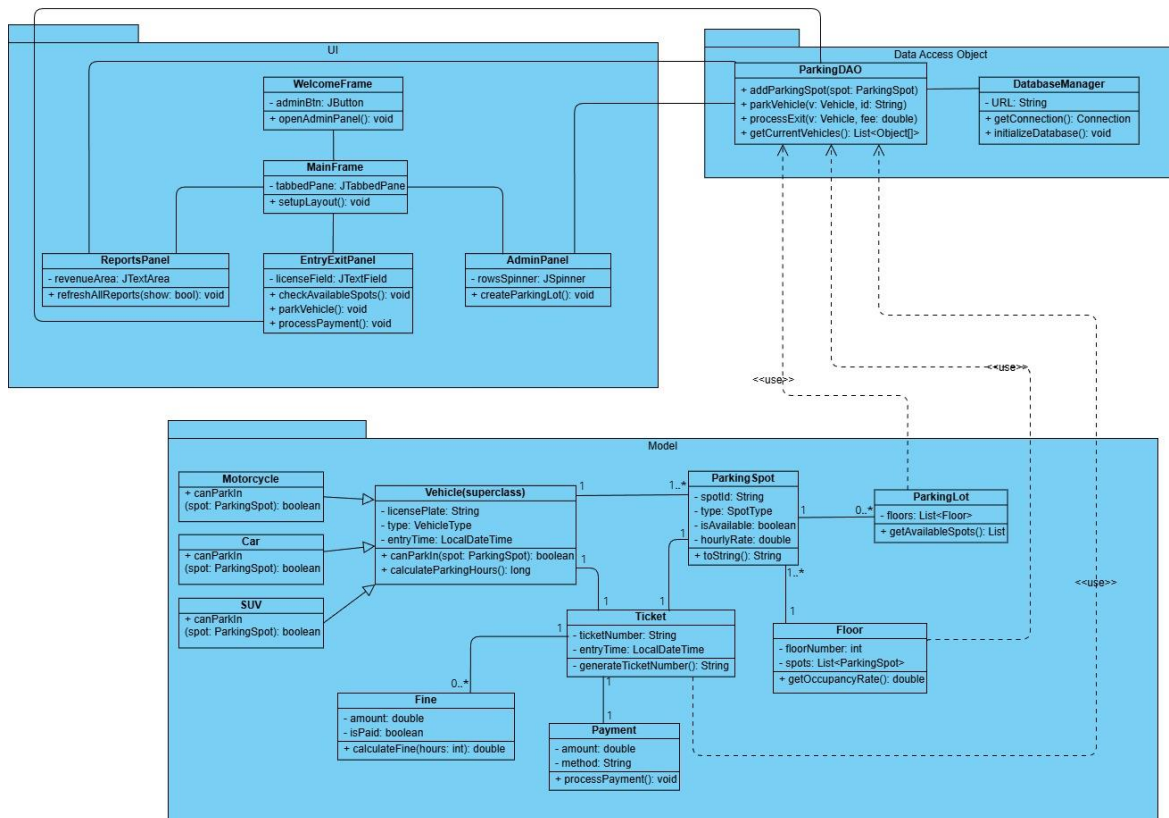
1. Use Case Diagram

The Use Case Diagram illustrates the functional requirements of the Parking Lot Management System by identifying the interactions between external actors and the system's internal processes.



2. Class Diagram

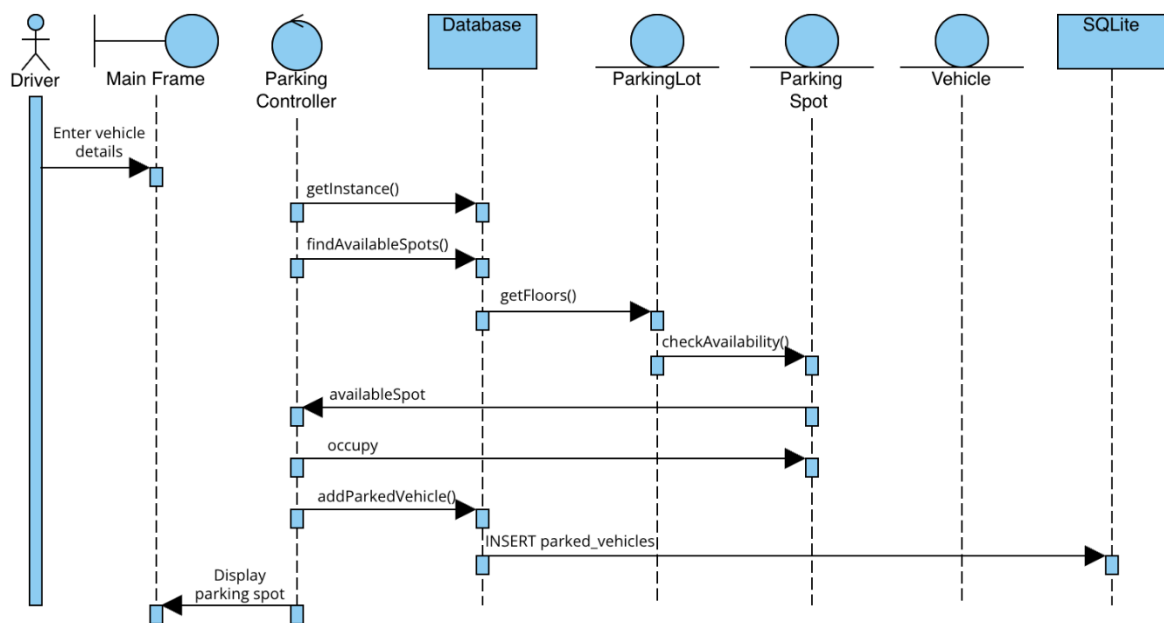
The system is designed using a layered architecture to ensure a clean separation between the user interface, the business logic, and the data storage. This design supports the DAO Pattern and the Singleton Pattern.



3. Sequence Diagram – Vehicle Parks

Flow:

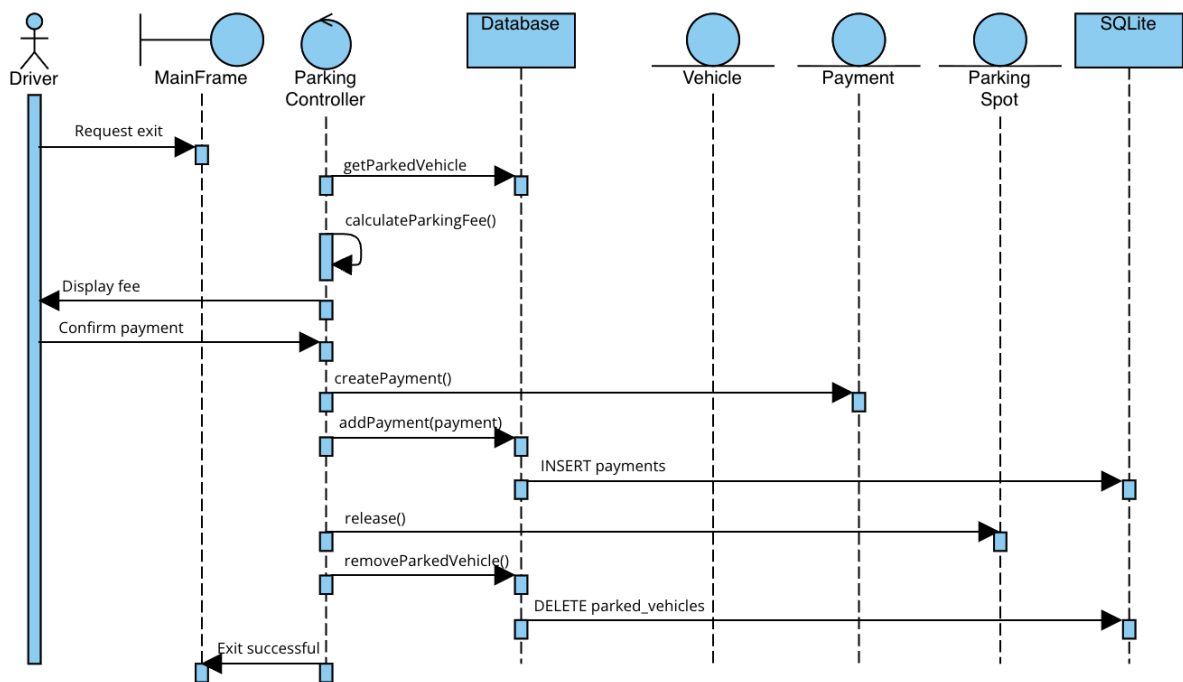
1. Driver enters vehicle details
2. GUI requests available spots
3. Database searches Parking Lot
4. Spot is allocated
5. Vehicle is stored in Database and SQLite
6. Entry time is recorded



4. Sequence Diagram – Vehicle Exits

Flow:

1. Driver requests exit
2. System calculates parking fee
3. Driver pays
4. Payment saved
5. Parking spot released
6. Vehicle removed from DB



5. Sequence Diagram – Vehicle Unpaid Fine

Flow:

1. Driver attempts exit
2. System detects unpaid fines
3. Exit is blocked
4. Driver pays fine
5. Fine status updated in DB
6. Exit allowed

